Enumerating Solid Sudoku Subcubes

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Abstract

In this paper, we enumerate and bound the Solid Sudoku Subcube (SSC), a cube of side length n restricted by the constraint that the integers 1 through n^2 must appear once in each slice. We establish that there is 1 SSC of order 2 and 448 SSCs of order 3, unique up to relabeling. Furthermore, using matrix permanents, we derive generalized combinatorial formulas that bound the total number of valid configurations for any order $n \geq 4$ SSC.

1 Introduction

A Sudoku grid is a 9×9 Latin square divided into nine 3×3 regions with the condition that every column, row and region is filled with the digits $1-9$, each of the digits appearing only once.

Three dimensionsional Sudoku has been constructed in a number of different ways. Church (2005) constructed a Sudoku Cube by layering nine Sudoku grids on top of each other with the added structure that three adjacent faces also formed Sudoku grids. Lambert & Whitlock (2010) generalized the Dion Cube so that all six faces could be used as Sudoku grids.

More recently, Solid Sudoku Cubes were introduced in Najafian *et al*. (2018). We discuss a particular subset of this object, the *Solid Sudoku Subcube* (SSC). An SSC is a cube of side length n constrained such that every integer from 1 to n^2 appears once per orthogonal slice.

The question of how many Sudoku grids there were intrigued mathematicians for decades and led to the solution found by Felgenhauer & Jarvis (2005). Russell & Jarvis

An interactive web implementation of SSCs is available at `https://cubiku.onrender.com`.

(2006) found the number of essentially different, nonequivalent Sudoku grids.

1.1 Definitions and Theorems

Definition 1.1. Let $A = (a_{ij})$ be an $m \times n$ matrix over any commutative ring with $m \leq n$. Let $P(m,n)$ denote the set of all one-to-one functions from $\{1,2,\ldots,m\}$ to $\{1,2,\ldots,n\}$. The *permanent* of A, per(A), is defined by

$$\text{per}(A) = \sum_{\sigma \in P(m,n)} a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{m,\sigma(m)}$$

Definition 1.2. Let X_1,X_2,...,X_m be subsets, not necessarily distinct, of an n-set $S = \{x_1, x_2, ..., x_n\}$. Let $A = (a_{ij})$ be the $(0,1)$-matrix whose entries are defined as follows:

$$a_{ij} = \begin{cases} 1 & \text{if} x_j \in X_i \\ 0 & \text{if} x_j \notin X_i \end{cases}$$

The matrix A is called the *incidence matrix* for the subsets X_1,X_2,...,X_m of the n-set S.

Definition 1.3. Let X_1,X_2,...,X_m be subsets of an n-set S. A sequence $(s_1, s_2, ..., s_m)$ of m distinct elements of S is said to form a *system of distinct representatives* (SDR) if $s_i \in X_i$, $i = 1, ..., m$.

(Minc (1984)) The number of SDRs for a set of sets with incidence matrix M is per(M).

Definition 1.4. Λ_n^k is the set of $n \times n$ $(0,1)$-matrices in which each row and column sum to k.

Theorem 1.1 (Minimum values of per(A), $A \in \Lambda_n^k$)**.** *(Wanless (2006)) For any integers $n \geq k \geq 1$ and any $A \in \Lambda_n^k$,*

$$\text{per}(A) > (\frac{(k-1)^{k-1}}{k^{k-2}})^n$$

Theorem 1.2 (Maximum values of per(A), $A \in \Lambda_n^k$)**.** *(Brègman (1973), Wanless (2003)) For $n = tk + r$ with $0 \leq r < k$*

$$(k!)^t r! \leq \max_{A \in \Lambda_n^k} \text{per}(A) \leq (k!)^{\frac{n}{k}}$$

The SSC is equivalent to the top n blocks (denoted $B_1, \ldots, B_n$) of an n-Sudoku grid, with the additional constraint that similar columns across these blocks must collectively contain the digits $\{1, \ldots, n^2\}$ once.

2 Combinatorial Enumeration

2.1 Counting SSCs of order 2

Exchanging any two digits in a valid grid results in another valid grid. Any grid can be relabeled so that the leftmost block (B_1) is given by

1	2		
3	4		

In this case we say the grid is in standard form, and our count reduces by $n!$. Our problem reduces to counting the number of cubes with B_1 in standard form.

Lemma 2.1. Up to standard-form relabeling, there exists a unique valid SSC configuration of order 2.

Proof. Assume B_1 is fixed in standard form. For the subsequent block B_2, the available candidates for any cell (r,c) are bounded by the elements already placed in the corresponding row and column of B_1. Since every digit must appear once in each slice, these cross-slice constraints leaves one valid digit placement per cell. Therefore, B_2 is uniquely determined. □

There is only one way to complete the grid that satisfies the constraints and so, with our ordered entries, we have 1 SSC of order 2 up to relabeling, and $4! \times 1 = 24$ valid 2-SSCs.

2.2 Counting SSCs of order 3

As before, we do relabeling on B_1, reducing our counting by $9! = 362,880$. We now have B_1 in standard form.

Determining possible entry combinations is essentially an SDR enumeration problem. Each of the nine cells in B_2 corresponds to a specific subset of the set $\{1,2,\ldots,9\}$, and we must pick distinct representatives from each of these subsets to form a valid block.

We therefore need to count the number of ways we can fill B_2. We have the following possibilities:

$\{5,6,8,9\}$	$\{4,6,7,9\}$	$\{4,5,7,8\}$
$\{2,3,8,9\}$	$\{1,3,7,9\}$	$\{1,2,7,8\}$
$\{2,3,5,6\}$	$\{1,3,4,6\}$	$\{1,2,4,5\}$

Table 1: B2 Entry Possibilities

We first form an incidence matrix for these 9 subsets. Define the subsets corresponding to the cells as follows:

$$X_1 = \{5,6,8,9\}, \qquad X_2 = \{4,6,7,9\}, \qquad X_3 = \{4,5,7,8\}$$
$$X_4 = \{2,3,8,9\}, \qquad X_5 = \{1,3,7,9\}, \qquad X_6 = \{1,2,7,8\}$$
$$X_7 = \{2,3,5,6\}, \qquad X_8 = \{1,3,4,6\}, \qquad X_9 = \{1,2,4,5\}$$

Construct the matrix A_1:

$$A_1 = (a_{ij}) = \begin{cases} 1 & \text{if} x_j \in X_i \\ 0 & \text{if} x_j \notin X_i \end{cases}$$

$$A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Proposition 2.2. For an order-3 SSC with block B_1 fixed in standard form, the incidence matrix A_1 of B_2 is regular, satisfying $A_1 \in \Lambda_9^4$. Furthermore, B_3 is completely determined by our choice for B_2.

Proof. For any empty cell (r,c) in B_2, the starting candidate pool is $3^2 = 9$. The corresponding row and column in B_1 restrict 3 digits each. To determine the overlap between these two restriction pools, we observe that the 3 digits in the row pool occupy

3 distinct columns in B_1. Because the slice rules distribute symbols uniformly, these 3 digits are partitioned equally across the 3 columns of the grid. Thus, only $\frac{3}{3} = 1$ digit overlaps with our target column c.

By the Principle of Inclusion-Exclusion, the number of excluded candidates is $3+3-1=5$. Subtracting this from the initial candidates pool leaves $9-5=4$ valid candidates per cell. Therefore, the incidence matrix A_1 for block B_2 maintains constant row and column sums of 4, meaning $A_1 \in \Lambda_9^4$.

For the final block B_3, the row and column constraints now span both B_1 and B_2. The row pool contains $2 \times 3 = 6$ restricted digits. Each of these 6 digits must appear once in each of the 2 completed blocks. Therefore, across B_1 and B_2, these 6 digits account for $2 \times 6 = 12$ total column placements. Distributing these placements uniformly across the 3 available columns gives an overlap of $\frac{12}{3} = 4$ digits shared with our target column c.

Applying inclusion-exclusion, the total excluded candidates are $6+6-4=8$. Subtracting this from the initial candidates pool leaves $9-8=1$ valid candidate per cell. This implies $A_2 \in \Lambda_9^1$. So B_3 is completely determined by our choice for B_2. □

Evaluating the incidence matrix A_1 gives $\mathrm{per}(A_1) = 448$, representing 448 valid completions for block B_2. As shown, this selection uniquely determines the final block B_3. Therefore, there are precisely 448 3-SSCs up to relabeling. Factoring in the 9! permutations for block B_1 gives a total of $448 \times 9! = 162,570,240$ valid 3-SSCs.

2.3 Counting SSCs of order n

Having established the combinatorial framework in the previous section, we now generalize these structural bounds to an arbitrary order n SSC.

Lemma 2.3. Suppose we have m validly completed blocks $(B_1, \ldots, B_m)$ of an n-SSC. For any cell (r,c) in the subsequent block B_{m+1}, the number of available candidates drops to $(n-m)^2$. The resulting incidence matrix A_m is regular, with $A_m \in \Lambda_{n^2}^{(n-m)^2}$.

Proof. Consider any empty cell (r,c) in B_{m+1}. Our starting pool of candidates is n^2.

The row passing through this cell spans the m preceding blocks, collecting a pool of mn restricted digits. The column passing through the cell does the same, collecting another pool of mn restricted digits. To find the remaining candidates, we must subtract these restrictions using the Principle of Inclusion-Exclusion, which requires us to find the exact number of digits shared between the row pool and the column pool.

Consider the mn distinct digits in the row pool. In the m completed blocks, each of these digits has been placed m times, occupying m distinct columns. Therefore,

across the m completed blocks, these mn digits account for $m \times mn = m^2 n$ total column placements.

Within the m completed blocks, each digit contributes m column placements. Due to the SSCs slice constraints, these $m^2 n$ total placements partition uniformly across the n available columns. As a result, the number of row pool digits intersecting target column c is $\frac{m^2 n}{n} = m^2$.

Applying inclusion-exclusion, the total number of excluded candidates is $2mn - m^2$. Subtracting this from our initial candidates pool gives the number of remaining candidates:

$$n^2 - (2mn - m^2) = n^2 - 2mn + m^2 = (n-m)^2$$

This holds identically for every cell in B_{m+1}. The reduction applies uniformly and the incidence matrix A_m maintains constant row and column sums of $(n-m)^2$. Thus, $A_m \in \Lambda_{n^2}^{(n-m)^2}$. □

Lemma 2.4. The total number of valid completions for an n-SSC is given by the product of the permanents of the incidence matrices $A_0, A_1, \ldots, A_{n-1}$. The overall count is bounded by the product of Brègman's upper bounds for each $\Lambda_{n^2}^{(n-m)^2}$.

Proof. We build an n-SSC sequentially, completing B_1, then B_2, all the way through B_n. At each step m, the number of valid ways to fill the subsequent block B_{m+1} corresponds to the number of valid matchings between the available positions and the remaining entries, which is measured by the permanent of our incidence matrix, $\mathrm{per}(A_m)$.

Because each block completion builds directly on the valid structure of the ones before it, the total number of full n-SSC configurations is the product of these permanents across all m steps. Applying Brègman's Theorem to the regular incidence matrices in $\Lambda_{n^2}^{(n-m)^2}$, each factor is bounded by its respective row-sum degree, giving us the upper bound for the entire space. □

We enumerate the order 4 SSCs by first fixing block B_1 in standard form and sequentially counting the remaining blocks. Just as in the $n = 3$ case, the slice constraints dictate that completing blocks $B_1, B_2,$ and B_3 leaves one valid assignment for each cell in the final block, meaning B_4 is completely determined.

We construct incidence matrices to count the completions for the blocks B_2 ($m = 1$) and B_3 ($m = 2$). By Lemma 2.3, the available candidates drops quadratically at each step, dropping from 9 candidates per cell in B_2 to 4 candidates per cell in B_3. The incidence matrices A_1 and A_2 fall into the classes Λ_{16}^9 and Λ_{16}^4, respectively.

To define the bounds on the total number of configurations x, we evaluate the permanent bounds of A_1 and A_2 together with the 16! permutations for block B_1. This

gives:

$$16! \times \left(\frac{8^8}{9^7}\right)^{16} \times \left(\frac{3^3}{4^2}\right)^{16} < x \leq 16! \times (9!)^{\frac{16}{9}} (4!)^{\frac{16}{4}}$$

The total number of valid configurations is bounded by:

$$4.75 \times 10^{25} < x \leq 5.31 \times 10^{28}$$

We carry this exact same logic over to order 5 SSCs. The sequential completion of blocks B_2, B_3, and B_4 gives the incidence matrices $A_1 \in \Lambda_{25}^{16}$, $A_2 \in \Lambda_{25}^{9}$, and $A_3 \in \Lambda_{25}^{4}$:

$$25! \times \left(\frac{15^{15}}{16^{14}}\right)^{25} \times \left(\frac{8^8}{9^7}\right)^{25} \times \left(\frac{3^3}{4^2}\right)^{25} < x \leq 25! \times (16!)^{\frac{25}{16}} (9!)^{\frac{25}{9}} (4!)^{\frac{25}{4}}$$

This gives the bound:

$$1.23 \times 10^{64} < x \leq 1.19 \times 10^{70}$$

More generally, for SSCs of order $n \geq 4$, the sequential completion of blocks B_2 through B_{n-1} is governed by incidence matrices $A_m \in \Lambda_{n^2}^{k_m}$, where the row-sum degree is defined by $k_m = (n-m)^2$ for $m = 1, \ldots, n-2$. Finding the permanents of these matrices gives the general bound:

$$(n^2)! \prod_{m=1}^{n-2} \left(\frac{(k_m - 1)^{k_m - 1}}{k_m^{k_m - 2}}\right)^{n^2} < x \leq (n^2)! \prod_{m=1}^{n-2} (k_m!)^{\frac{n^2}{k_m}}$$

3 Future Work

These SSC structures can be embedded onto a physical n-Rubik's cube. This approach differs from standard novelty Sudoku cubes in that the spatial center is treated mathematically as an ordinary node. These mappings can be explored to determine the number of possible solvable puzzle states a given SSC can generate.

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